



Dr. Alfonso Ortega

James R. Birle Professor of Energy Technology

Associate Vice President for Research and Graduate Programs

Site Leader, NSF I/UCRC in Energy-Efficient Electronic Systems

Villanova University

Dr. Alfonso Ortega is the James R. Birle (pronounced “Burr Lee”) Professor of Energy Technology and Associate Vice President for Research and Graduate Programs at Villanova University. He is the Director of the Laboratory for Advanced Thermal and Fluid Systems which he founded in 2005. He is the Site Director for the NSF Center for Energy Smart Electronic Systems (ES2) founded in 2011 with Binghamton University, University of Texas-Arlington, and Georgia Tech.

He received his B.S. from The University of Texas-El Paso, and his M.S. and Ph.D. from Stanford University, all in Mechanical Engineering. Dr. Ortega was on the faculty of the Department of Aerospace and Mechanical Engineering at The University of Arizona in Tucson for 18 years, where he directed the Experimental and Computational Heat Transfer Laboratory. From 2004 to 2006, Dr. Ortega was the Program Director for Thermal Transport and Thermal Processing in the Chemical and Transport Systems Division of The National Science Foundation in Arlington, Virginia, where he managed the NSF’s primary program funding heat transfer and thermal technology research in U.S. universities.

Dr. Ortega is an internationally recognized researcher in the areas of thermal management of data centers and electronic systems, convective and conjugate heat transfer in complex flows, experimental measurements in the thermal sciences, and thermal management in energy systems. He has supervised over 40 M.S. and Ph.D. candidates to degree completion, 5 postdoctoral researchers, and more than 70 undergraduate research students. He is the author of over 300 journal and symposia papers, book chapters, and monographs.

Dr. Ortega is a Fellow of the ASME. He is the recipient of the National Science Foundation Presidential Young Investigator Award, the ASME Electronic Packaging Division Thermal Management Award, the SEMITHERM Significant Contributor Award, and the Harvey Rosten Award for Excellence. He has been an Associate Editor of the *ASME Journal of Heat Transfer*, Associate Editor of the *ASME Journal of Electronic Packaging* and Guest Associate Editor of the *IEEE Transactions on Components, Packaging, and Manufacturing Technology*. He has chaired the ASME K16 Committee on Heat Transfer in Electronic Equipment, the ASME Electronic and Photonic Packaging Division, the ASME/IEEE ITherm Symposium, the IEEE SEMITHERM Symposium, and the ASME InterPACK Conference.

June 2015.